

Model name: 4_25ep_medparam_4xconv2d_dense128

Description and underlying idea:

As a strategy for the first Convolutional Layer we use medically relevant kernels:

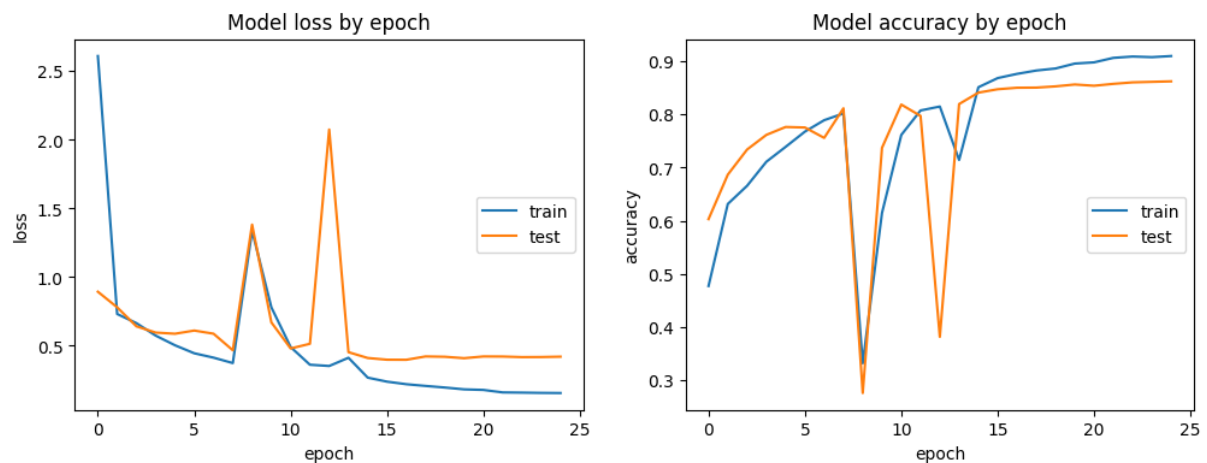
- sobel filters → detect lung boundaries
- laplacian filters → highlight nodules and opacities
- gabor filters → capture texture changes (useful for pneumonia, fibrosis)
- gaussian filters → enhance image contrast

apply small noise and transformations to all filters to ensure kernel diversity

Conclusion:

The training history shows, that there is still potential for further learning. In the next model 4_50ep_medparam_4xconv2d_dense128 we have let the model further trained. However, the prediction performance is already not too bad. Also the presetting with medical convolutional kernels seemed to had not really an effect on the performance.

Training History:



Performance Report / Recall normalized confusion matrix

	pre	rec	f1
COVID	0.934337	0.957924	0.945983
Lung_Opacity	0.798748	0.766524	0.782304
Normal	0.864839	0.873664	0.869229
Viral Pneumonia	0.908197	0.932660	0.920266
avg_pre	0.861398	0.861398	0.861398
avg_rec	0.862509	0.862509	0.862509
avg_spe	0.914208	0.914208	0.914208
avg_f1	0.861815	0.861815	0.861815

